

HS 151: Economics

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Description of the course

The course will primarily deal with introductory Economics concepts, both from Microeconomics and Macroeconomics, specifically designed for students on to a non-Economics career path. From Microeconomics, it will include topics such as markets, firms and consumer behaviours. From Macroeconomics, it will include topics such as national income, economic growth, unemployment, inflation, money and banking, and international trade. The aim will be to focus on general interest questions, such as, “Should we always make in India” or “Why can’t we keep printing money”. It will also cater to Engineering students by helping them understand the role of Economics in their profession.

Learning Outcomes

The course is designed to achieve the following learning outcomes for students:

- Students will learn to **the key Economics concepts**.
- They will **begin to understand** how consumers, producers and governments come together to **produce and consume** for personal and for common good
- They will also get a **primer on setting up business**, and **managing personal finances**.

Most importantly, as Engineers, they will see how technologies and innovations interact with Economic lives.

Texts/References:

Textbooks

- Principles of Microeconomics, Karl E. Case, Ray C. Fair, Sharon M. Oster
- The Economy: A South Asian Perspective. *coreecon*.
- N Gregory Mankiw. *Principles of Economics*.

Other ‘General Economics’ books

- Economics for the Common Good, Jean Tirole
- The Undercover Economist, Tim Harford
- Good Economics for Hard Times, Abhijit Banerjee and Esther Duflo
- *More to be added*

Grading Scheme (TBA)

Course Outline (*Note: there might be some changes during the course*)

1. Introduction (The big picture): Why should we care about Economics?
 - a. How do economists view the world?
 - b. Importance of Economics in our daily lives
2. Using economics for individual decision making, and strategic interactions
 - a. The model of Individual choices (Scarcity, choice, MRS, MRT)
 - b. Strategic interactions (self-interest, social interest, altruism, game theory)
3. How to think like an Economist for setting up and running a business/firm?
 - a. Understanding the market (Demand-Supply) – quantity and prices in Competitive versus Monopoly cases
 - b. Understanding firm's vitals: revenue, profit and cost (sunk/variable/fixed)
 - c. Role of location in setting up prices. Where to set up shop? Why do similar shops cluster in the same place, and the role of information cost?
4. Viewing a country as an Economist (Part-I): Does a country really need a government?
 - a. Free market versus market failures (Externalities - positive and negative)
 - Environmental externalities
 - b. Public goods vs. private goods; the problem of coordination
 - c. Role of taxation, subsidies in redistribution
 - d. Solution to Environmental problems
5. Viewing a country as an Economist (Part-III): Is it enough if India becomes a 5 trillion economy?
 - a. Growth and its measures: GDP, per capita income
 - b. Development and its measures: inequality, poverty, the concept of environment
6. Viewing a country as an Economist (Part-II): What is the role of money in an economy, and why can't we keep printing it if it's that important?
 - a. Money, its meaning, its functions
 - b. Prices in the context of inflation/deflation
 - c. Time value of money
 - d. The monetary system. Role of commercial banks and central banks.
7. Economics of managing personal finances
 - a. Finance for a non-finance person?

- b. The power of compounding
 - c. Investment and instruments of investments
 - d. Is just saving enough?
8. Viewing the world as an Economist. Should we always “make in India”?
- a. Trade – benefits and downsides (absolute advantage versus comparative advantage)
 - b. International trade – import duty, export promotion, exchange rate, appreciation, depreciation.
9. Economics of Innovation and Technology: Will AI take away all the jobs?
- a. Understanding unemployment
 - b. Creative Destruction
 - c. AI Economics
 - d. Production possibilities and technology

DISCLAIMERS:

- Changes to this syllabus may occur during the semester at the discretion of the instructor.
- Students will themselves be responsible to stay updated with all the announcements made in the class/tutorials and Google Classrooms.
- Given the size of the class, please do not expect responses over emails.
- The following activities may result in direct reporting to SSAC.
 - o Sharing the links to in-class quizzes with those who are absent on the day of the quiz.
 - o Taking in-class quizzes without being in the class.
 - o Providing any biometric records without attending the class.
 - o All those mentioned on the SSAC website:
<https://iitgn.ac.in/students/honourcode>